

Face Recognition and Verification via Transfer Learning

Fine-Tuning InceptionResNetV1 for Face Classification and Building an Embedding-Based Face Verifier

Alireza Amini

github.com/AlirezaAmini81/Face-Recognition-and-Verification-Project

1. Face Classifier

First, pre-trained InceptionResNetV1 model is loaded and a simple two-layer MLP is attached to the output of its embedding layer, following the architecture shown in Figure 1. Several training configurations are then evaluated below. In addition, different learning rates together with two optimizers (SGD and Adam) were tested for the first configuration, and the best performance was obtained with Adam at a learning rate of 0.0001.

```
(fc1): Linear(in_features=512, out_features=512, bias=True)
(relu): ReLU()
(fc2): Linear(in_features=512, out_features=14, bias=True)
```

Figure 1: Architecture of the two-layer MLP head

1.1 Case 1

In this configuration, all layers of the InceptionResNetV1 network are frozen, and gradients are propagated only into the MLP head.

As shown after 20 epochs, the lowest test loss reached is 2.46 and the highest test accuracy is 95.19%. Next, the effect of unfreezing all layers except the last one is examined.

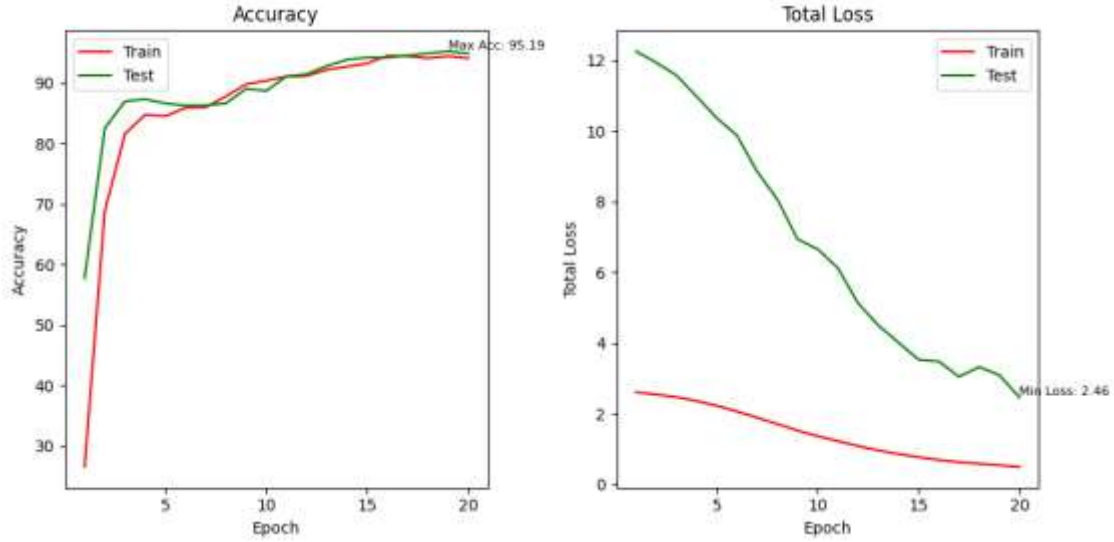


Figure 2: Accuracy and loss curves for Case 1

1.2 Case 2

In this configuration, all layers of InceptionResNetV1 are frozen except its last layer, and gradients flow into the MLP head as well as the final layer of InceptionResNetV1.

Compared with the previous configuration, at the same number of epochs the test accuracy is higher, reaching 96.22%, although the loss value increased slightly. This can be attributed to the fact that only one layer of the feature extractor was unfrozen; unfreezing a larger number of layers may allow the network to learn better features for the classifier.

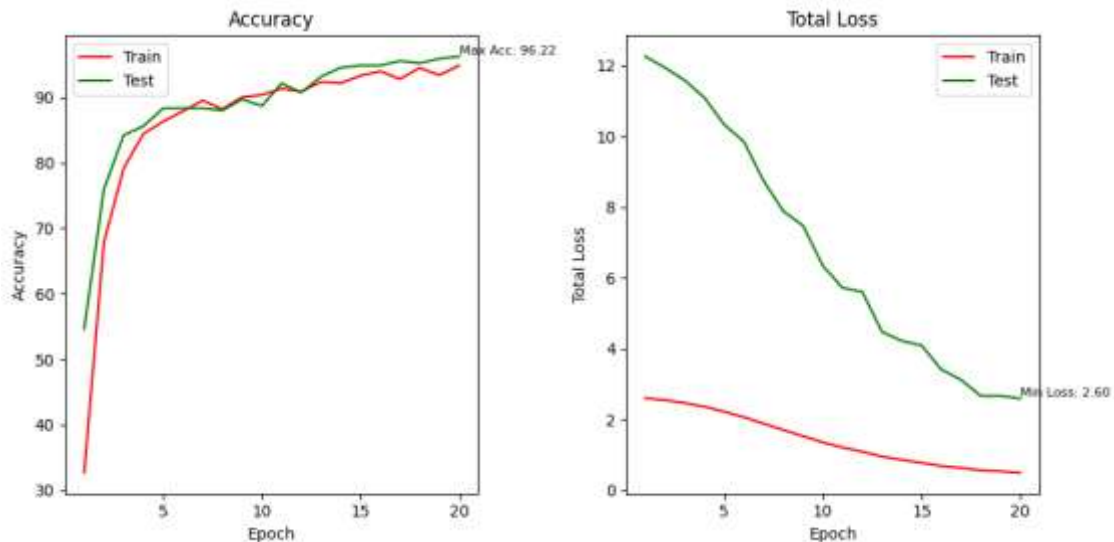


Figure 3: Accuracy and loss curves for Case 2

1.3 Case 3

In this configuration, all layers of InceptionResNetV1 are frozen except its last 5 layers, and gradients flow into the MLP head as well as these 5 layers of InceptionResNetV1.

As shown, the lowest test loss is 0.51 and the highest test accuracy is 97.94%. As expected, unfreezing more layers allows the feature extractor to derive better class-discriminative features, which considerably reduces the loss.

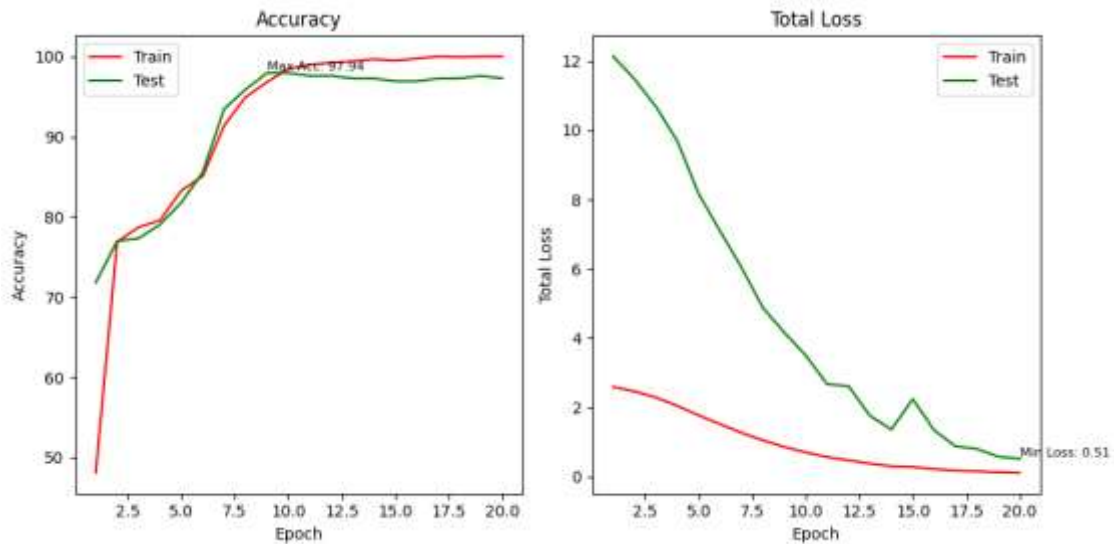


Figure 4: Accuracy and loss curves for Case 3

1.4 Case 4

In this configuration, all layers of InceptionResNetV1 are frozen except its last 10 layers, and gradients flow into the MLP head as well as these 10 layers of InceptionResNetV1.

This configuration achieves the best results overall: the highest accuracy of 99.66% and the lowest test loss of 0.32. It is therefore selected as the best model.

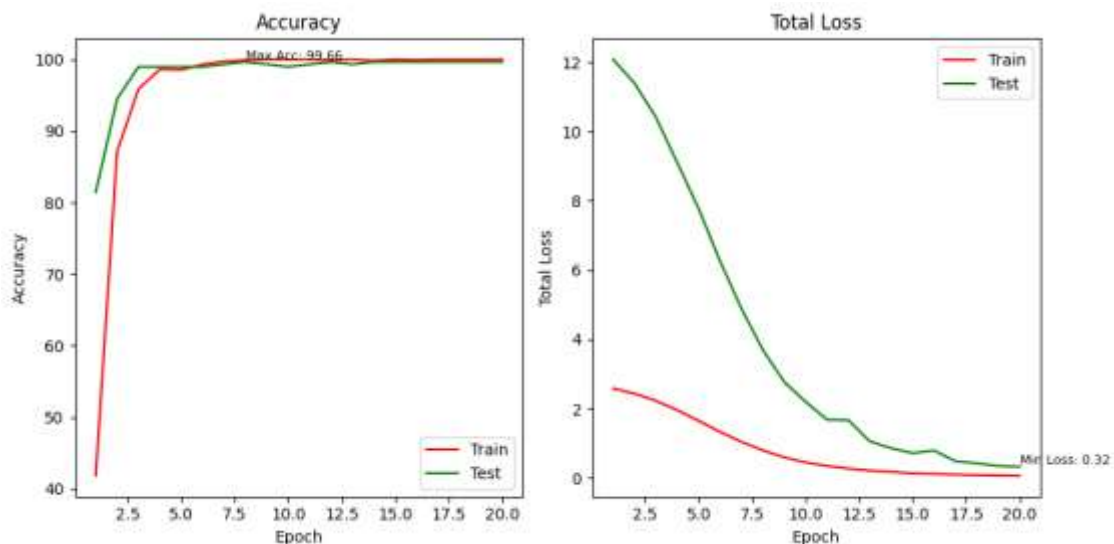


Figure 5: Accuracy and loss curves for Case 4

Since this configuration produced the best accuracy, a deeper three-layer MLP head with dropout and batch normalization was also tested on top of the same feature extractor, using the architecture shown below.

```
(fc1): Linear(in_features=512, out_features=512, bias=True)
(relu1): ReLU()
(batch_norm1): BatchNorm1d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
(drop_out1): Dropout(p=0.3, inplace=False)
(fc2): Linear(in_features=512, out_features=256, bias=True)
(relu2): ReLU()
(batch_norm2): BatchNorm1d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
(fc3): Linear(in_features=256, out_features=14, bias=True)
```

Figure 6: Architecture of the three-layer MLP head

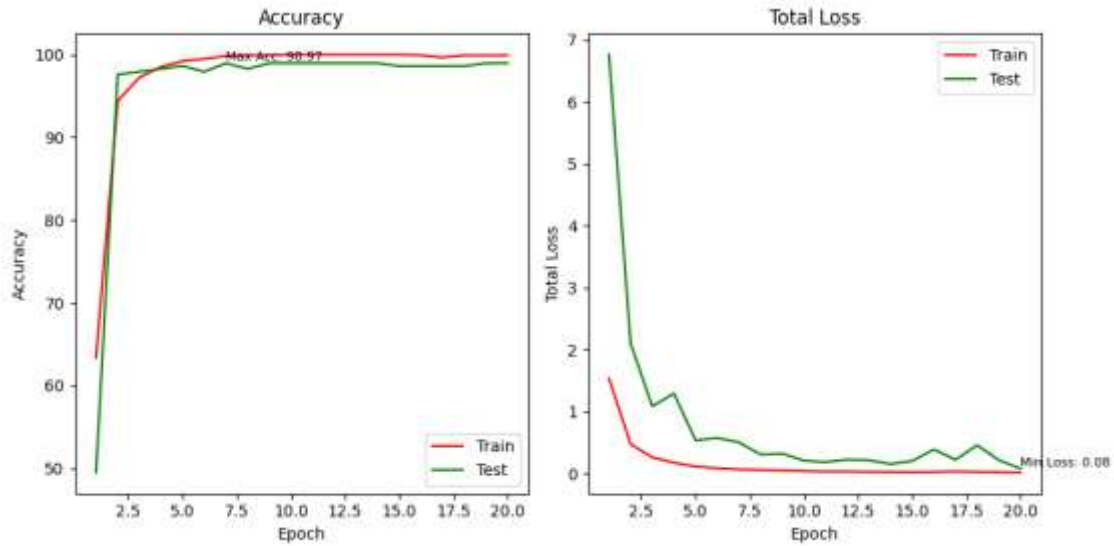


Figure 7: Accuracy and loss curves for the three-layer MLP head

As shown, although the loss decreases noticeably compared with the previous (two-layer) MLP head, the test accuracy is lower. Consequently, the two-layer MLP model is chosen for the next part.

2. Face Verifier

In this part, the embedding layer is first used to extract 512-dimensional feature vectors for all of the test data. 95% of these vectors are then assigned to a set of “known” features, and the remaining data is assigned to a set of “unknown” features.

Since the best model selected in the previous phase reached an accuracy of 99.66%, when the cosine similarity and the Euclidean distance between each unknown feature vector and all known feature vectors are computed, both metrics perform equally well and correctly return a sample from the same class for every query.

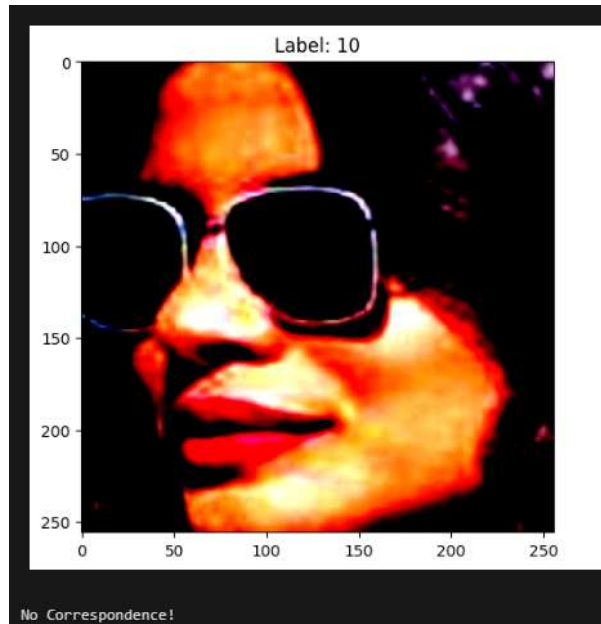


Figure 8: An example of a non-match case

To better evaluate the two similarity/distance metrics, 70% of the data is assigned to the known set and the remaining 30% to the unknown set, in order to have more data available for evaluation. In addition, a query is declared “no match” if the cosine similarity is below 0.9, or if the Euclidean distance is above 0.002.

Finally, the accuracy of both methods is measured based on the percentage of queries for which a same-class sample was returned. This accuracy was 97.7272% for cosine similarity and 98.8636% for Euclidean distance.

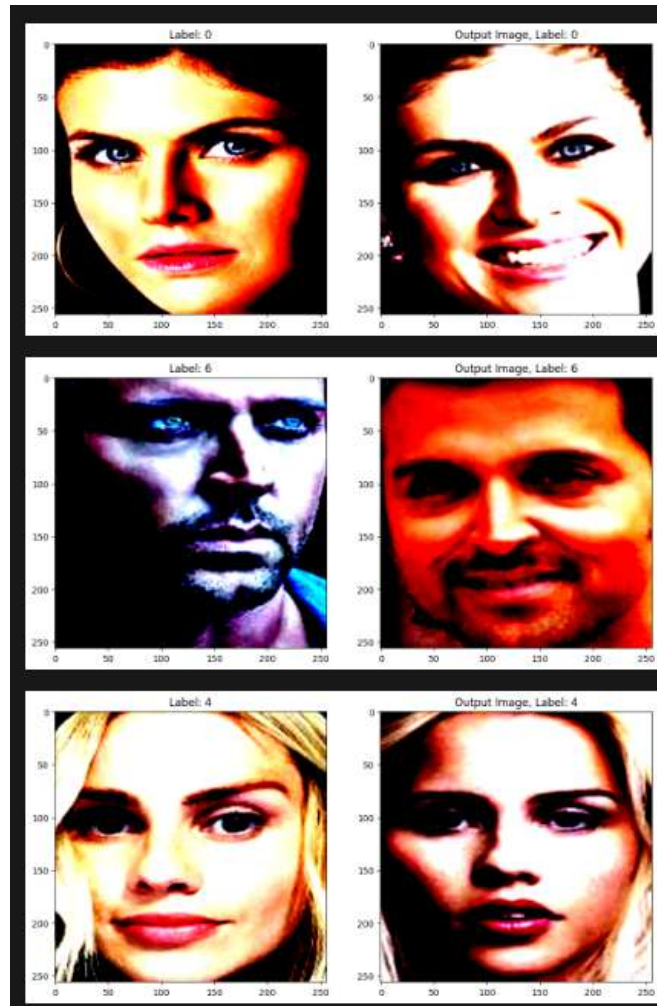


Figure 9: Examples of correct recognition